MILITARY SPECIFICATION

RESISTOR, FIXED, COMPOSITION (INSULATED), ESTABLISHED RELIABILITY

STYLE RCR42

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RCR42 resistor.

2. APPLICABLE DOCUMENT

2.1 Government specification. Unless otherwise specified, the following specification, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-39008 - Resistors, Fixed, Composition (Insulated), Established Reliability, General Specification for.

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-39008, and as specified herein.

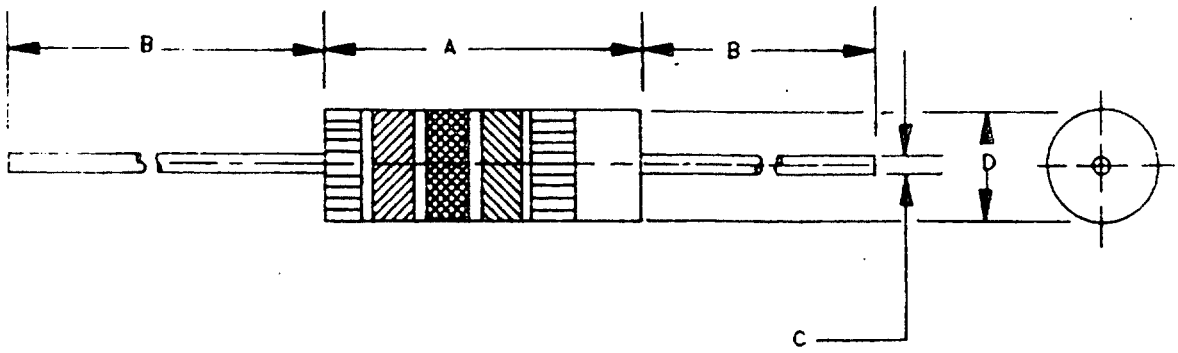
3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be 2 watts, based on full load operation (100 percent rated wattage) at an ambient temperature of 70°C. Resistors operated at ambient temperature in excess of 70°C shall be derated in accordance with figure 2. The rated dc or rms continuous working voltage shall not exceed 500 volts.

3.4 Resistance. The minimum resistance value shall be 10 ohms; maximum value shall be 22 megohms.

3.5 Weight. The maximum weight of this style shall be 3.0 grams.

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Ltr	Dimensions
A	.688 (17.48) ±.040 (1.02)
B	1.500 (38.10) ±.125 (3.18)
C	.045 (1.14) ±.003 (.08)
D	.318 (8.08) ±.018 (.46)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only

FIGURE 1. Style RCR42 resistor.

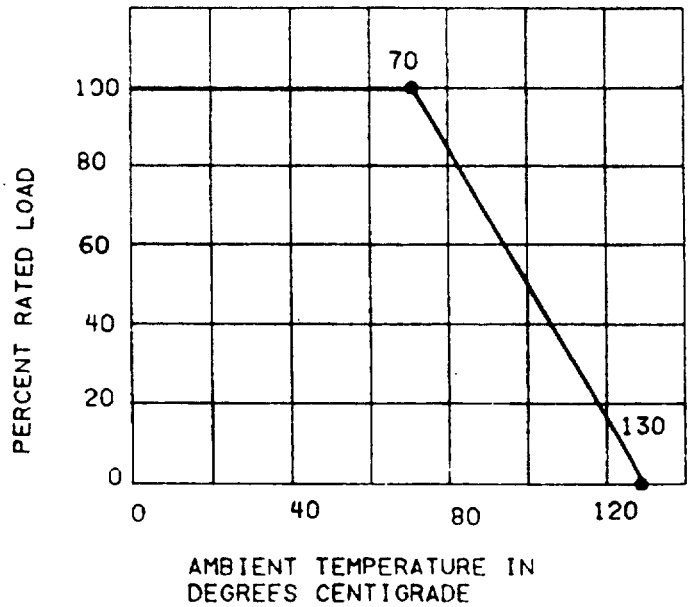


FIGURE 2. Derating curve for high ambient temperatures.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-39008 and as specified herein.

*5. PACKAGING

*5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39008.

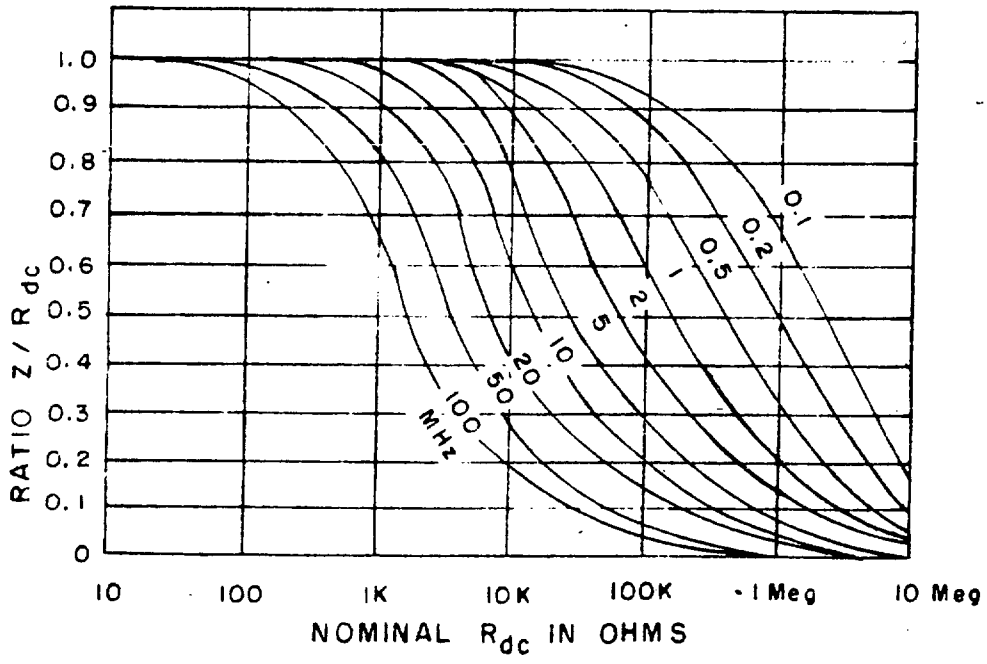
6. NOTES

6.1 Notes. The notes specified in MIL-R-39008 are applicable to this specification.

6.2 Power rating. It is essential that no more than 50 percent of the rated wattage be applied if the established failure rate level is to be maintained.

6.3 MIL-R-11 substitution data. Resistors of this specification, regardless of their failure rate designation, are substitutes for resistors of the same resistance value and tolerance specified in MIL-R-11/7.

*6.4 High frequency characteristics. Typical values of impedance to dc resistance ratio and phase angle from 100 kHz to 100 MHz are shown on figures 3 and 4 below. Circuit variations in mounting position and lead length can have a significant effect on the high frequency characteristics.



2 WATT TYPE RCR42

FIGURE 3. Impedance to dc resistance ratio.

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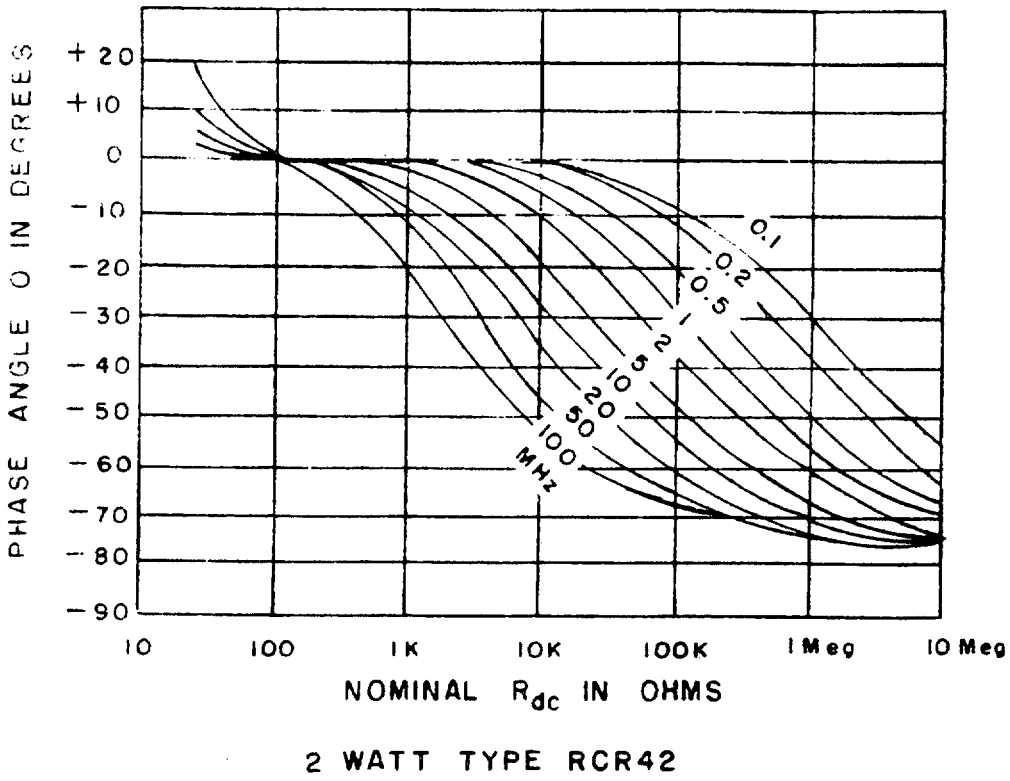


FIGURE 4. Impedance to phase angle.

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* 6.5 Peak voltage and pulsed operation. When these resistors are used under low duty cycle pulse applications, the maximum permissible operating voltage is limited by breakdown instead of heating. In such applications, the peak value of the pulse shall not exceed 2.5 times rated rms working voltage or 1,000 volts, whichever is less.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - ER, AR, MI
Navy - EC, AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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